Klausur SS09

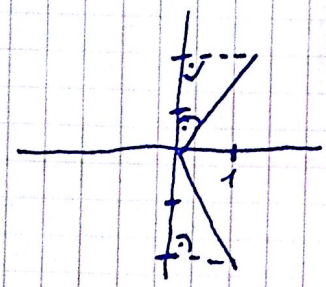
Aufgabe 1

a) $z = 2-4i \quad |z| = \sqrt{4+16} = \sqrt{20}$

$\varphi = \arctan\left(\frac{-4}{2}\right) + 2\pi = 296{,}56°$

b) $z = 2 \cdot e^{i\frac{1}{3}\pi} = 2 \cdot (\cos(60°) + i \cdot \sin(60°)) = 1 + 1{,}73i$

c) $z = \frac{(1+2i)\cdot(1+2i)}{(1-2i)\cdot(1+2i)} = \frac{1+4i+4i^2}{5} = -\frac{3}{5} + \frac{4}{5}i$



$r = \sqrt{5}, \quad \varphi_1 = 63{,}43°$

$\varphi_2 = 296{,}56°$

$z = \frac{\sqrt{5} \cdot e^{i\,63{,}43°}}{\sqrt{5} \cdot e^{i\,296{,}56°}} = e^{i\,-233{,}13°}$

$\left.\begin{array}{l} x = \cos(-233{,}13°) = -0{,}6 \\ y = \sin(-233{,}13°) = 0{,}8 \end{array}\right\} z = -\frac{3}{5} + \frac{4}{5}i$

d) $z = 1+i = \sqrt{2} \cdot e^{i45°}; \; n = 3, \; a_0 = \sqrt{2}, \; \alpha = 45°$

$z_k = \sqrt[3]{\sqrt{2}} \cdot \left(\cos\left(\frac{45° + k \cdot 360°}{3}\right) + i \cdot \sin\left(\frac{45° + k \cdot 360°}{3}\right)\right)$

$z_0 = 1{,}122(\cos 15° + i \cdot \sin 15°) = 1{,}08 + 0{,}29i$

$z_1 = 1{,}122(\cos 135° + i \cdot \sin 135°) = -0{,}79 + 0{,}79i$

$z_2 = 1{,}122(\cos 255° + i \cdot \sin 255°) = -0{,}29 - 1{,}08i$

2. a) $1 + \sum_{k=1}^{n} \frac{2^{2(k-1)}}{3^k} = \frac{4^n}{3^n}$ für $n \in \mathbb{N} \wedge n \geq 1$

R.S.: $\frac{4^{n+1}}{3^{n+1}} = \frac{4 \cdot 4^n}{3 \cdot 3^n} = \frac{4}{3} \cdot \frac{4^n}{3^n}$ ✓

L.S.: $\frac{4^n}{3^n} + \frac{2^{2n}}{3^{n+1}} = \frac{4^n}{3^n} + \frac{1 \cdot 4^n}{3 \cdot 3^n} = \frac{4}{3} \cdot \frac{4^n}{3^n}$ ✓

b) $n^2 - 2n - 1 > 0$ für $n \in \mathbb{N} \wedge n \geq 3$

$(n+1)^2 - 2(n+1) - 1 > 0 \Leftrightarrow n^2 + \cancel{2n} + 1 - \cancel{2n} - 2 - 1 > 0$

$n^2 - 2 > 0 \Leftrightarrow n^2 > 2 \Rightarrow$ wahr da $n \geq 3$

c) $\sum_{r=1}^{n} (r \cdot r!) = (n+1)! - 1$ für $n \in \mathbb{N} \wedge n \geq 1$

R.S.: $(n+2)! - 1$ =

L.S.: $(n+1)! - 1 + (n+1) \cdot (n+1)!$

$\cancel{= n! + 1 - 1 + (n+1) \cdot (n! + 1)}$

$\cancel{= n! + n \cdot n! + n + n! + 1}$

$= (n+1)! \, (1 + (n+1) - 1 = (n+1)! \cdot (n+2) - 1 = (n+2)! - 1$ ✓